

Learning without Forgetting (LwF)

Project Documentation and Experimental Results

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Abstract

Deep neural networks can suffer from catastrophic forgetting: performance on a previously learned task may deteriorate when the model is adapted to a new task. This project reproduces the central idea of Learning without Forgetting (LwF) by Li and Hoiem. An ImageNet-pretrained AlexNet model is adapted to two new classification tasks: CUB-200-2011 bird classification and MIT Indoor Scenes classification. Three approaches are compared: Fine-Tuning, Feature Extraction, and LwF. The implementation uses a consistent experimental protocol for both scenarios, including a one-epoch warm-up stage, validation-based checkpoint selection, fixed old-task evaluation on 80 ImageNet validation batches, and a single fixed random seed. The reproduced results follow the main qualitative trend of the original paper: Fine-Tuning causes the strongest forgetting, Feature Extraction preserves the old task most effectively but adapts less strongly to the new task, and LwF provides a favorable trade-off between retention and adaptation.

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1 Introduction

Deep neural networks are often trained under the assumption that all relevant training data remain available whenever the model is adapted. In practice, this assumption may be unrealistic. Original data may be unavailable because of storage limitations, privacy restrictions, licensing constraints, or computational cost.

When a model is trained on a new task without access to the original training data, its previously learned capabilities may degrade. This phenomenon is known as catastrophic forgetting. The paper Learning without Forgetting (LwF) proposes a practical approach for learning a new task while preserving performance on an old task, without requiring access to the old task’s training images.

The key idea is to use the original model as a frozen teacher. During adaptation, the adapted model is trained both to classify the new-task data correctly and to preserve the teacher’s old-task responses on those same new-task images. This combines transfer learning with knowledge distillation.

This project reproduces the main LwF behavior in two scenarios:

- ImageNet $\rightarrow$ CUB-200-2011
- ImageNet $\rightarrow$ MIT Indoor Scenes

The same evaluation and training protocol is used in both experiments to ensure a clear and reproducible comparison.

2 Project Objectives

The objectives of this project are:

- Reproduce the central Learning without Forgetting framework.
- Compare LwF with Fine-Tuning and Feature Extraction.
- Measure the effect of each method on both the old and new tasks.
- Evaluate whether the reproduced results follow the qualitative trends reported in the original paper.
- Propose a possible improvement to the original LwF method.

3 Compared Methods

3.1 Fine-Tuning

Fine-Tuning adapts a pretrained model by optimizing the shared network representation and the newly added task-specific classifier using new-task data.

In this project, the old ImageNet head is retained and frozen for evaluation. Therefore, after Fine-Tuning, the same old head can be used to measure how changes in the shared representation affected ImageNet performance.

Advantage: the network can strongly adapt its representation to the new task.

Disadvantage: because the shared parameters are optimized only for the new task, old-task performance can decline significantly.

3.2 Feature Extraction

Feature Extraction freezes the pretrained representation and trains only a new classifier for the new task. In this implementation, dropout is disabled inside the frozen extractor, and a two-layer new-task head is trained:

$$4096 \rightarrow 4096 \rightarrow N_{\text{new}},$$

where N_{new} is the number of classes in CUB or Scenes.

Advantage: the original ImageNet representation is preserved exactly.

Disadvantage: the shared representation cannot adapt to features that are particularly useful for the new task.

3.3 Learning without Forgetting (LwF)

LwF combines new-task classification with old-task response preservation. The implementation uses:

- a frozen ImageNet teacher model;
- a dual-head student model;
- an old ImageNet head with 1000 outputs;
- a new-task head with 200 outputs for CUB or 67 outputs for Scenes.

The total training loss is:

$$L = L_{\text{new}} + \lambda_{\text{old}} L_{\text{old}},$$

where L_{new} is the standard cross-entropy classification loss for the new task, and L_{old} is a distillation loss that encourages the student’s old-task responses to remain close to the teacher’s responses.

For temperature T , the softened probability distribution is:

$$p_i^{(T)} = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}.$$

The response-preservation loss is implemented as:

$$L_{\text{old}} = T^2 \cdot D_{\text{KL}} \left(p_{\text{teacher}}^{(T)} \parallel p_{\text{student}}^{(T)} \right).$$

In both experiments:

$$T = 2, \quad \lambda_{\text{old}} = 1.$$

4 Experimental Setup

4.1 Model Architecture

Both reproduced experiments use AlexNet pretrained on ImageNet. This is consistent with the main experimental setting of the original LwF paper.

For Fine-Tuning and LwF, the model is represented as:

- shared convolutional layers;
- shared fully connected classifier layers;
- old ImageNet classification head;
- new-task classification head.

4.2 Datasets

4.2.1 ImageNet

ImageNet is the original task. The old-task head predicts one of 1000 object classes. The original ImageNet training data are not used during adaptation. ImageNet validation images are used only after training to measure old-task retention.

4.2.2 CUB-200-2011

CUB-200-2011 is a fine-grained bird-classification dataset containing 200 bird species. The official train/test split is preserved. An internal validation subset is created only from the official training set.

4.2.3 MIT Indoor Scenes

MIT Indoor Scenes is an indoor-scene classification dataset with 67 categories. The official training set contains 5360 images and the official test set contains 1340 images. An internal validation subset is created only from the official training set.

4.3 Consistent Reproduction Protocol

The CUB and Scenes notebooks use the same protocol. Table 1 summarizes the settings.

Table 1: Consistent protocol used in both reproduced experiments.

Setting	Value
Architecture	ImageNet-pretrained AlexNet
Number of completed runs	1
Random seed	42
Warm-up stage	1 epoch
Fine-Tuning stage	5 epochs
Feature Extraction stage	5 epochs
LwF stage	5 epochs
Internal validation fraction	15% of the official new-task training set
ImageNet evaluation	First 80 validation batches
ImageNet batch size	64
ImageNet DataLoader shuffling	Disabled (shuffle=False)
New-task evaluation	Complete official test set
Base learning rate	0.001
Head-only learning rate	0.005
Feature Extraction learning rate	0.005
Momentum	0.9
Weight decay	0.0005
Distillation temperature	$T = 2$
Old-task loss coefficient	$\lambda_{\text{old}} = 1$

4.4 Warm-Up and Checkpoint Selection

Before Fine-Tuning and LwF, the new-task head is trained for one warm-up epoch while the pretrained representation and old ImageNet head remain frozen. Fine-Tuning and LwF then start from the same warmed-up checkpoint. This provides a fairer comparison and more closely follows the procedure described in the original paper.

For each new-task dataset, the official training set is divided into an internal training subset and a held-out validation subset. The checkpoint with the best validation accuracy is selected during training. The official CUB and Scenes test sets are reserved exclusively for final evaluation and are not used for checkpoint selection.

4.5 ImageNet Evaluation Constraint

Due to runtime and hardware constraints, old-task ImageNet accuracy is evaluated on the first 80 validation batches rather than the complete validation set:

$$80 \times 64 = 5120 \text{ images.}$$

The same fixed evaluation protocol is applied to the ImageNet baseline, Fine-Tuning, Feature Extraction, and LwF in both experiments. This makes comparisons within this project meaningful. However, the resulting ImageNet values should not be interpreted as full-validation-set benchmark results.

4.6 Data Preprocessing

For the new-task training subsets, the implementation uses:

- resize to 256×256 pixels;
- random crop to 224×224 pixels;
- random horizontal flip;
- ImageNet normalization.

Validation and test images use deterministic center crops. This reduces overfitting and creates a cleaner separation between training and evaluation.

5 Reproduction Process

The reproduction process is executed separately for the CUB and Scenes datasets.

First, an ImageNet-pretrained AlexNet model is loaded and evaluated on the fixed 80-batch ImageNet validation subset. This establishes the old-task baseline.

Second, a dual-head model is created. The original ImageNet head is retained, and a randomly initialized new-task head is added. The new head is trained for one warm-up epoch while all other parameters remain frozen.

Third, Fine-Tuning and LwF are initialized from the same warmed-up state:

- **Fine-Tuning:** the shared representation and new-task head are optimized using only the new-task classification loss. The old head remains frozen and is later used to evaluate forgetting.
- **LwF:** the shared representation, old head, and new head are jointly optimized using both the new-task classification loss and the old-task distillation loss.

Feature Extraction is trained separately using the frozen pretrained representation and a two-layer new-task head.

Finally, the selected checkpoint for each method is evaluated on the complete official new-task test set and on the first 80 ImageNet validation batches.

6 Reproduced Results

6.1 Absolute Performance

Table 2 reports the absolute results obtained by the notebooks. The baseline ImageNet value is identical in both scenarios because both use the same pretrained model and the same fixed ImageNet evaluation subset.

Table 2: Absolute performance of the reproduced experiments.

Experiment	Method	Old Task Accuracy (ImageNet)	New Task Accuracy
ImageNet $\rightarrow$ CUB	LwF	59.49	45.41
ImageNet $\rightarrow$ CUB	Fine-Tuning	53.42	45.53
ImageNet $\rightarrow$ CUB	Feature Extraction	62.89	42.03
ImageNet $\rightarrow$ Scenes	LwF	59.82	56.27
ImageNet $\rightarrow$ Scenes	Fine-Tuning	48.69	50.82
ImageNet $\rightarrow$ Scenes	Feature Extraction	62.89	55.00

6.2 Presentation in the Original Paper’s Style

The original paper reports the absolute performance of LwF and expresses the other methods as differences relative to LwF:

$$\Delta = \text{Method Accuracy} - \text{LwF Accuracy}.$$

Positive values indicate that a method outperformed LwF on a specific task, while negative values indicate lower performance.

Table 3: Reproduced results in the presentation style of the original paper. LwF is reported in absolute values; the other rows report differences relative to LwF.

Method	ImageNet $\rightarrow$ CUB		ImageNet $\rightarrow$ Scenes	
	Old	New	Old	New
LwF (ours)	59.49	45.41	59.82	56.27
Fine-Tuning	-6.07	+0.12	-11.13	-5.45
Feature Extraction	+3.40	-3.38	+3.07	-1.27

7 Comparison with the Original Paper

Table 4 shows the corresponding AlexNet results from the original paper. As in Table 3, the LwF row contains absolute values, and the remaining rows contain differences relative to LwF.

Table 4: Original paper results for the corresponding AlexNet experiments.

Method	ImageNet $\rightarrow$ CUB		ImageNet $\rightarrow$ Scenes	
	Old	New	Old	New
LwF (ours)	54.7	57.7	55.9	64.5
Fine-Tuning	-3.8	-0.7	-2.0	-0.8
Feature Extraction	+2.3	-5.2	+1.2	-3.3

7.1 ImageNet $\rightarrow$ CUB Analysis

The reproduced CUB results follow the main retention-adaptation trade-off:

- Feature Extraction preserves the old ImageNet task best: 62.89%.
- Fine-Tuning suffers from stronger forgetting: 53.42% on ImageNet.
- LwF retains substantially more old-task performance than Fine-Tuning: 59.49% on ImageNet.
- On the new CUB task, LwF and Fine-Tuning are effectively tied: 45.41% and 45.53%, respectively.
- Feature Extraction is weaker on the new CUB task: 42.03%.

The original paper reports a small new-task advantage for LwF over Fine-Tuning. In this reproduction, Fine-Tuning exceeds LwF by only 0.12 percentage points on CUB. Because the experiment was executed once with a single fixed seed, this very small difference should not be interpreted as strong evidence that Fine-Tuning is generally superior. The central trend remains consistent: LwF preserves old-task performance far better than Fine-Tuning while learning the new task substantially better than Feature Extraction.

7.2 ImageNet $\rightarrow$ Scenes Analysis

The Scenes results match the qualitative ordering reported in the original paper:

- Feature Extraction preserves ImageNet most effectively: 62.89%.
- Fine-Tuning causes the strongest forgetting: 48.69% on ImageNet.
- LwF retains most of the old-task knowledge: 59.82% on ImageNet.
- LwF achieves the strongest new-task performance: 56.27% on Scenes.
- Feature Extraction is slightly weaker on Scenes: 55.00%.
- Fine-Tuning is weaker on both tasks in this run: 50.82% on Scenes.

This experiment clearly illustrates the intended role of LwF: it allows the shared representation to adapt while constraining old-task drift.

8 Bonus Optimization Experiment

In addition to reproducing the main LwF experiments, we performed a bonus optimization experiment on the **ImageNet $\rightarrow$ MIT Indoor Scenes** scenario. The goal of this experiment was to improve the reproduced LwF result in a controlled way, without changing the dataset, evaluation protocol, ImageNet evaluation subset, temperature value, or old-task loss coefficient.

The optimization focuses on the new-task classification head. Instead of using a standard linear classifier for the MIT Indoor Scenes head, the optimized model uses a **scaled cosine-normalized classifier**. This modification was designed to improve the new-task decision boundary while keeping the original LwF distillation mechanism responsible for preserving ImageNet knowledge.

8.1 Optimization Motivation

In the standard LwF setup, the model is trained using the combined loss:

$$L = L_{\text{new}} + \lambda_{\text{old}} L_{\text{old}},$$

where L_{new} is the cross-entropy loss for the new Scenes task, and L_{old} is the distillation loss that encourages the student model to preserve the teacher model’s ImageNet responses.

This structure already provides a strong stability-plasticity trade-off. However, the new Scenes classifier is randomly initialized and must learn a 67-class classification problem from a relatively small dataset. Therefore, improving the new-task head is a focused way to improve performance without aggressively changing the shared representation.

8.2 Scaled Cosine-Normalized Classifier

In a standard linear classifier, the class logit is computed as a dot product:

$$z = f \cdot w,$$

where f is the feature vector and w is the class weight vector. This score depends on both the direction and the magnitude of the vectors. In transfer learning, this can make the classifier sensitive to weight-norm differences rather than only to semantic similarity.

The optimized classifier normalizes both the feature vector and the class weights:

$$z = \eta \cdot \frac{f \cdot w}{\|f\| \|w\|}.$$

This makes the classifier depend mainly on angular similarity. As a result, class decisions are based on how well the image representation matches each class direction, rather than on uncontrolled growth of feature or weight magnitudes.

Because cosine similarity is naturally limited to the range $[-1, 1]$, the classifier uses a learnable scale parameter η . In the final optimized run, the scale was initialized as:

$$\eta_0 = 30.$$

This produces stronger logits for the softmax layer and helps the new Scenes classifier learn more effectively. During training, the scale remained stable and reached:

$$\eta = 30.26.$$

8.3 Boosted Learning Rate for the New Task Head

The second part of the optimization was increasing the learning rate only for the new Scenes head:

$$LR_{\text{new head}} = 5 \times LR_{\text{base}}.$$

The shared backbone and the old ImageNet head continued to use the regular base learning rate. This is important because the new head needs fast adaptation, while the shared representation should remain stable enough to avoid catastrophic forgetting.

This design improves plasticity where it is needed most, in the new classifier, while keeping stability in the shared layers through the LwF distillation loss.

8.4 Optimization Results

Table 5 compares the standard LwF result with the optimized LwF result.

Table 5: Final optimization improvement over standard LwF for ImageNet $\rightarrow$ Scenes.

Method	ImageNet Old	Scenes New	Old Improvement	New Improvement
Standard LwF	59.82%	56.27%	–	–
Optimized LwF	60.72%	58.06%	+0.90%	+1.79%

The optimized method improved the MIT Indoor Scenes accuracy from:

$$56.27\% \rightarrow 58.06\%,$$

which is an improvement of:

$$+1.79\%.$$

At the same time, the ImageNet old-task accuracy improved from:

$$59.82\% \rightarrow 60.72\%,$$

which is an improvement of:

$$+0.90\%.$$

This is the main strength of the optimization: it improved the new-task performance without sacrificing old-task preservation. In continual learning, this is especially important because improvements on the new task often come at the cost of catastrophic forgetting on the old task.

8.5 Comparison with the Reproduced Methods

Table 6 compares the optimized method with the reproduced Fine-Tuning, Feature Extraction, and standard LwF results.

Table 6: Comparison between reproduced methods and the optimized LwF method.

Method	ImageNet Old	Scenes New	Main Observation
Fine-Tuning	48.69%	50.82%	Significant forgetting
Feature Extraction	62.89%	55.00%	Strong retention, weaker adaptation
Standard LwF	59.82%	56.27%	Balanced baseline
Optimized LwF	60.72%	58.06%	Best new-task accuracy with strong retention

Compared to Fine-Tuning, the optimized LwF method achieved both higher Scenes accuracy and much stronger ImageNet preservation. Fine-Tuning reached only 50.82% on Scenes and dropped ImageNet accuracy to 48.69%, while the optimized LwF method reached 58.06% on Scenes and preserved ImageNet accuracy at 60.72%.

Compared to Feature Extraction, the optimized LwF method achieved better new-task adaptation. Feature Extraction preserved ImageNet well, but reached only 55.00% on Scenes because the shared representation was not adapted enough to the new task. The optimized LwF method achieved higher Scenes accuracy while still maintaining strong ImageNet performance.

8.6 Why the Optimization Works

The optimization is effective because it improves the part of the network that is most responsible for learning the new task: the new Scenes classification head.

The cosine-normalized classifier creates a more balanced decision boundary by comparing classes through angular similarity rather than raw vector magnitude. The learnable scale parameter makes the logits strong enough for effective softmax classification. The boosted learning rate allows the new head to adapt faster to MIT Indoor Scenes, while the LwF distillation loss continues to protect the old ImageNet behavior.

Overall, the optimized method improves the balance between stability and plasticity. It gives the model more flexibility where it is needed, in the new task head, while preserving the old-task knowledge through the original LwF mechanism.

8.7 Optimization Summary

The final optimized method, **LwF with a scaled cosine classifier and boosted new-head learning rate**, improved the reproduced ImageNet $\rightarrow$ Scenes results.

Compared to standard LwF, it improved the MIT Indoor Scenes accuracy by +1.79% and the ImageNet old-task accuracy by +0.90%. The method does not require additional data, does not change the evaluation protocol, and does not require a larger backbone model. The improvement comes from a focused architectural and training modification that makes the new-task classifier more suitable for continual learning.

9 Differences from a Full Replication

This project is a computationally constrained reproduction of the main LwF trend rather than a full replication of every experiment in the original paper.

- **Single run:** each experiment was executed once with seed 42. The original paper reports averaged results across multiple runs.
- **ImageNet subset:** old-task performance was measured on the first 80 ImageNet validation batches rather than the entire validation set.
- **Subset ordering:** because `shuffle=False`, the first 80 batches form a fixed but not necessarily representative subset. This may affect absolute ImageNet values.

- **Training budget:** each method used five primary training epochs, preceded by a one-epoch warm-up for Fine-Tuning and LwF.
- **Augmentation differences:** the implementation uses random crop and horizontal flip. The original paper also applies additional augmentation details, including RGB variation.
- **Scope:** this project focuses on Fine-Tuning, Feature Extraction, and LwF. Other baselines from the original paper, such as Joint Training, LFL, and Fine-Tune FC, were outside the project scope.
- **Implementation framework:** the reproduction uses PyTorch, while the original paper used MatConvNet.

Despite these differences, both experiments reproduce the main conceptual trade-off between preserving old capabilities and adapting to a new task.

10 Discussion

The experiments demonstrate the core challenge of continual learning. Fine-Tuning can adapt the model to a new task, but the shared representation may drift away from features needed by the original task. This leads to catastrophic forgetting.

Feature Extraction represents the opposite extreme. Because the pretrained representation remains frozen, ImageNet performance is preserved. However, the model has less flexibility to adapt its internal representation to CUB or Scenes.

LwF provides a compromise. The shared representation remains trainable, but the distillation loss discourages changes that strongly alter old-task behavior. The Scenes experiment demonstrates this trade-off clearly: LwF achieves the best Scenes accuracy while retaining much more ImageNet performance than Fine-Tuning. In the CUB experiment, LwF and Fine-Tuning are essentially tied on the new task, while LwF preserves substantially more ImageNet accuracy.

The results also emphasize the importance of fair evaluation. The notebooks use the same AlexNet architecture, warm-up stage, number of epochs, random seed, checkpoint-selection method, and fixed ImageNet subset in both scenarios. This consistency supports a fair comparison between the two reproduced scenarios.

11 Conclusion

This project reproduced the main framework of Learning without Forgetting and evaluated it on ImageNet $\rightarrow$ CUB and ImageNet $\rightarrow$ Scenes scenarios.

The implementation uses AlexNet in both experiments, a one-epoch warm-up stage, a single fixed seed, five epochs per compared method, validation-based checkpoint selection, full official new-task test evaluation, and a shared 80-batch ImageNet evaluation protocol.

The reproduced results support the main conclusion of the original paper. Fine-Tuning causes the strongest forgetting. Feature Extraction preserves old-task performance most effectively but limits adaptation. LwF provides a favorable balance by retaining more old-task

knowledge than Fine-Tuning while learning the new task more effectively than Feature Extraction. On Scenes, LwF achieves the strongest new-task result. On CUB, LwF and Fine-Tuning are nearly tied on the new task, while LwF preserves considerably more ImageNet performance.

As a bonus extension, this project also tested an optimized LwF variant for the ImageNet $\rightarrow$ Scenes scenario using a scaled cosine classifier and a boosted new-head learning rate. This optimization improved the Scenes result while also maintaining strong ImageNet old-task performance.

12 References

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